

42.3 Piecewise polynomials

Changing the basis left the second complaint untouched: polynomials are global, and a degree-16 fit spends the same sixteen degrees of freedom everywhere, so a feature at one end is paid for by an oscillation at the other. The remedy is a different space of functions: cut the range at a few points, fit a low-degree polynomial on each piece, and join the pieces smoothly. Nothing oscillates, and the result is still linear in its coefficients.

42.3.1 Knots and pieces

Definition 42.3.1 — Piecewise polynomials and continuity constraints

Fix an interval $[a, b]$ and **knots** $a = \kappa_0 < \kappa_1 < \dots < \kappa_K < \kappa_{K+1} = b$, the K inner ones being the **interior knots**. A function $f : [a, b] \rightarrow \mathbb{R}$ is a **piecewise polynomial of degree d** if its restriction to each half-open piece $[\kappa_k, \kappa_{k+1})$, $k = 0, \dots, K-1$, and to $[\kappa_K, b]$ agrees with a polynomial of degree at most d ; the value at a knot is the limit from the right, and nothing ties the pieces together. These form a vector space $\mathcal{PP}_d(\kappa)$ of dimension $(K+1)(d+1)$.

For $-1 \leq m \leq d-1$, f satisfies the **continuity constraints of order m** at κ_k if

$$f^{(r)}(\kappa_k^-) = f^{(r)}(\kappa_k^+), \quad r = 0, 1, \dots, m,$$

the case $m = -1$ imposing nothing. The **spline space $\mathcal{S}_d(\kappa)$** is the set of piecewise polynomials of degree d satisfying the constraints of order $d-1$ at every interior knot; its members are the **splines of degree d with knots κ** .

The constraints are linear in the coefficients: writing the unrestricted piecewise polynomial as $\sum_k \sum_{j \leq d} c_{kj} (t - \kappa_k)^j \mathbf{1}_{\{t \in [\kappa_k, \kappa_{k+1})\}}$, the requirement $f^{(r)}(\kappa_k^-) = f^{(r)}(\kappa_k^+)$ reads

$$\sum_{j \geq r} \frac{j!}{(j-r)!} c_{k-1,j} (\kappa_k - \kappa_{k-1})^{j-r} = r! c_{kr},$$

one equation for each r and each interior knot. Fitting under them is restricted least squares (Proposition 11.3.3), and comparing two levels of smoothness is a general linear hypothesis (Theorem 11.3.2). Counting equations gives a candidate dimension, $(K+1)(d+1)$ coefficients minus $(m+1)K$ constraints, which for $m = d-1$ is

$$(K+1)(d+1) - dK = d+1 + K.$$

Only a candidate, since it assumes the constraints independent: the next theorem proves they are, by exhibiting a basis of exactly that size.

42.3.2 The spline space and the truncated power basis

Write $(v)_+ = \max(v, 0)$, and for $d \geq 1$ let

$$(t - \kappa)_+^d = \begin{cases} (t - \kappa)^d, & t \geq \kappa, \\ 0, & t < \kappa, \end{cases}$$

the **truncated power function** of degree d with knot κ . For $d = 0$ the same formula with the convention $0^0 = 1$ gives the indicator of $[\kappa, \infty)$.

Theorem 42.3.1 — Dimension of the spline space and the truncated power basis

Let $d \geq 0$ and let $\kappa_1 < \dots < \kappa_K$ be interior knots of $[a, b]$.

(a) $\mathcal{S}_d(\kappa)$ is a vector space of dimension $d + 1 + K$, and

$$1, t, \dots, t^d, (t - \kappa_1)_+^d, \dots, (t - \kappa_K)_+^d \quad (42.3.1)$$

is a basis of it.

(b) More generally, if the constraints imposed at κ_k are of order m_k with $-1 \leq m_k \leq d - 1$, the resulting space has dimension $d + 1 + \sum_{k=1}^K (d - m_k)$, with basis the powers $1, \dots, t^d$ together with $(t - \kappa_k)_+^r$ for $r = m_k + 1, \dots, d$ and $k = 1, \dots, K$.

Proof. It is enough to prove (b); part (a) is the case $m_k = d - 1$ for every k , for which the count is $d + 1 + K$.

The space. Sums and scalar multiples of piecewise polynomials of degree d are piecewise polynomials of degree d , and the constraints are linear equations, so the set is a subspace of the space of functions on $[a, b]$.

The listed functions belong to it. Each power t^j is a single polynomial, hence in every such space. Fix k and $r > m_k$. The function $g(t) = (t - \kappa_k)_+^r$ is zero on $[a, \kappa_k)$ and $(t - \kappa_k)^r$ on $[\kappa_k, b]$, a piecewise polynomial of degree $r \leq d$ whose only possible breakpoint is κ_k . For $s \leq r - 1$ its s th derivative at κ_k is $\{r!/(r - s)!\}(t - \kappa_k)^{r-s}$ there, which is zero, and zero from the left; so g satisfies the constraints of order $r - 1$, in particular those of order $m_k \leq r - 1$. At the other knots g is a single polynomial and satisfies any constraint.

Linear independence. Suppose

$$q(t) + \sum_{k=1}^K \sum_{r > m_k} c_{kr} (t - \kappa_k)_+^r = 0 \quad \text{for all } t \in [a, b],$$

with $q \in \mathcal{P}_d$. On $[a, \kappa_1)$ every truncated power vanishes, so q vanishes on an interval and hence $q \equiv 0$. Suppose inductively that $c_{lr} = 0$ for every $l < k$. On $[\kappa_k, \kappa_{k+1})$ the displayed sum reduces to $\sum_{r > m_k} c_{kr} (t - \kappa_k)^r$, a polynomial vanishing on an interval, so all its coefficients are zero. The induction gives $c_{kr} = 0$ throughout.

Spanning. Let f belong to the space. On $[\kappa_0, \kappa_1)$ it agrees with a polynomial $q \in \mathcal{P}_d$;

set $f_1 = f - q$, again in the space and vanishing there. Suppose inductively that f_k is in the space and vanishes on $[a, \kappa_k)$, and on the next piece agrees with $r_k \in \mathcal{P}_d$. Expanding r_k in powers of $t - \kappa_k$,

$$r_k(t) = \sum_{s=0}^d \frac{r_k^{(s)}(\kappa_k)}{s!} (t - \kappa_k)^s,$$

and for $s \leq m_k$ the constraint at κ_k forces $r_k^{(s)}(\kappa_k) = f_k^{(s)}(\kappa_k^-) = 0$, because f_k is identically zero to the left. Hence $r_k(t) = \sum_{s>m_k} c_{ks}(t - \kappa_k)^s$ for some coefficients c_{ks} , and $f_{k+1} = f_k - \sum_{s>m_k} c_{ks}(t - \kappa_k)^s_+$ is in the space and vanishes on $[a, \kappa_{k+1})$: to the left of κ_k both terms vanish, and between κ_k and κ_{k+1} they cancel. After K steps f_{K+1} vanishes on all of $[a, b]$, which exhibits f as a combination of the listed functions.

Dimension. The listed functions are $d+1 + \sum_k (d - m_k)$ in number, and they are a basis. ■

Remark. This settles the counting question: $(K+1)(d+1)$ minus $d+1 + \sum_k (d - m_k)$ is $\sum_k (m_k + 1)$, exactly the number of constraint equations. The constraints are therefore independent, and that a free piecewise polynomial is in fact a spline is testable with $\sum_k (m_k + 1)$ degrees of freedom (Definition 11.3.1).

42.3.3 Three familiar special cases

Degree 0. $\mathcal{S}_0(\kappa)$ is the space of step functions, of dimension $K+1$, and fitting it is the one-way layout with the intervals as groups (Example (The one-way layout)): the fit is *exactly local*, with variance σ^2/n_k on interval k , at the price of the discontinuity and a bias of order the interval width.

Degree 1. $\mathcal{S}_1(\kappa)$, of dimension $K+2$, is the space of continuous broken lines, with basis $1, t, (t - \kappa_1)_+, \dots, (t - \kappa_K)_+$; the coefficient of $(t - \kappa_k)_+$ is the *change* in slope there, and the two-piece case is the classical two-phase regression.

Degree 3. $\mathcal{S}_3(\kappa)$ is the space of **cubic splines**, of dimension $K+4$: value, slope and curvature continuous, only the third derivative jumping. Cubic is the default, the lowest degree whose joins are invisible and whose second derivative a smoothness penalty can control.

Warning. A knot estimated from the data takes the model out of this chapter: $(t - \kappa)_+^d$ is not linear in κ , the criterion is not even differentiable in it for $d = 0$ or 1 , and the usual asymptotics can fail. This **segmented** or **change point** regression has a literature of its own — Hudson (1966), Feder (1975), Seber and Wild (1989, ch. 9) — which this book does not develop. The honest options are to fix the knot from outside the data, to fit many fixed knots and let a penalty decide (Chapter 43), or to pay for the selection (Proposition 29.5.1).

42.3.4 Climbing the smoothness ladder

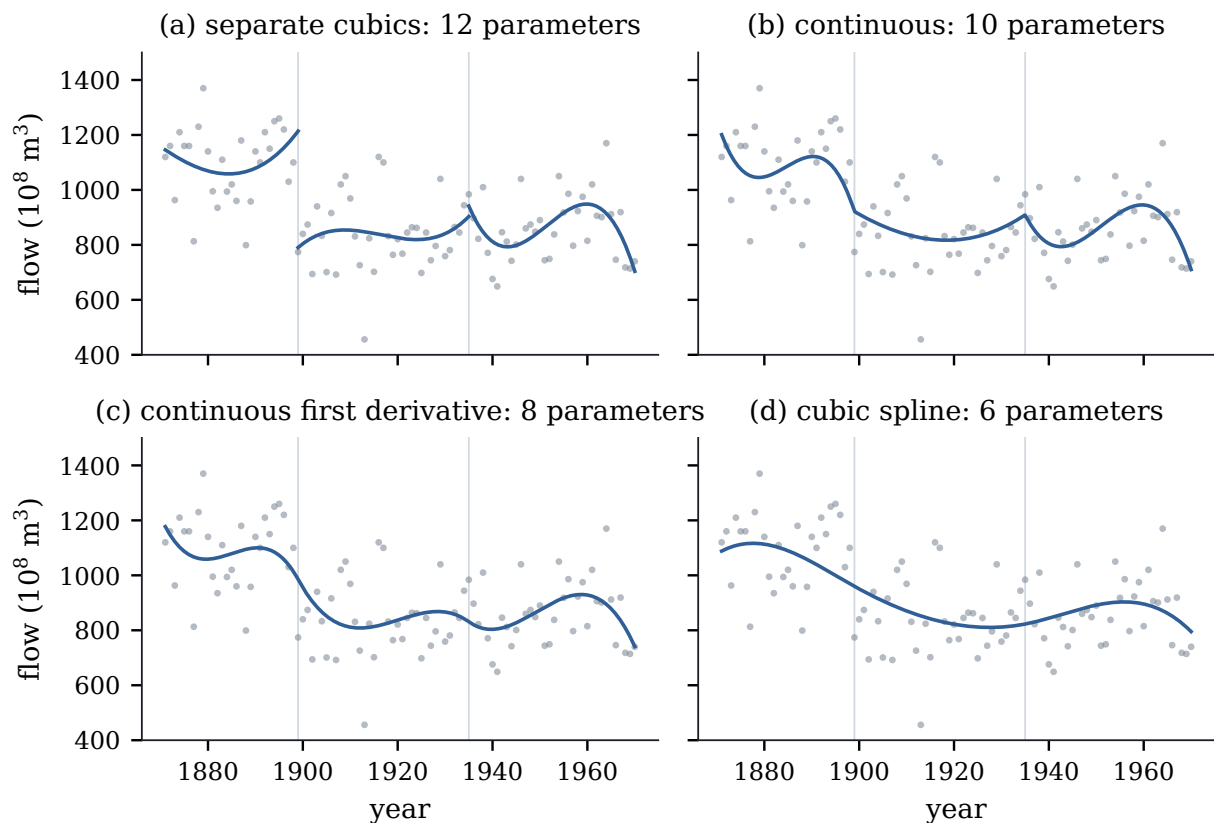


Figure 42.3.1: The Nile flows with cubic pieces cut at 1899 and 1935, under four levels of continuity; each constraint removes two parameters.

Example. (Four fits with the same knots)

Take the Nile series again, $n = 100$ annual flows, with cubic pieces and interior knots at 1899 — where Cobb (1978) and the analysis of Example (A trend in the Nile) both locate a change — and 1935. Fitting the three cubics separately uses 12 parameters and leaves a residual sum of squares of 1394072. Continuity at both knots removes two parameters and raises it to 1561452; continuity of the first derivative removes two more (1613581); and the full cubic spline, with 6 parameters as [Theorem 42.3.1](#) predicts, has 1781797. [Figure 42.3.1](#) shows the four fits.

Each step down the ladder is a testable linear hypothesis, and against the unrestricted fit all three are rejected: $F = 5.28$ on $(2, 88)$ degrees of freedom for continuity alone ($p = 0.0068$), $F = 3.46$ on $(4, 88)$ ($p = 0.0112$) for a continuous first derivative, and $F = 4.08$ on $(6, 88)$ ($p = 0.0012$) for the cubic spline. The Nile really does have a discontinuity, and a spline is the wrong space for it.

The right model is the simplest on the ladder: a step function with a single knot at 1899, whose group means are 1097.7 and 850.0 in units of 10^8 m^3 , differing by -247.8 with standard error 28.4 and $t = -8.71$. Its residual sum of squares, 1597457, beats the six-parameter spline and the cubic polynomial's 1909955. Two parameters in the right place beat six in the wrong space.

```
import numpy as np
import statsmodels.api as sm
```

```

nile = sm.datasets.nile.load_pandas().data
year = nile["year"].to_numpy().astype(float)
y = nile["volume"].to_numpy()
n = len(y)
knots = np.array([1899.0, 1935.0])
d = 3
u = lambda s: 2 * (s - year.min()) / (year.max() - year.min()) - 1

def separate(s):
    """One free cubic per interval: (K+1)(d+1) columns, no constraints at the knots.
    """
    piece = np.digitize(s, knots)
    return np.column_stack([np.where(piece == k, u(s) ** j, 0.0)
                           for k in range(len(knots) + 1) for j in range(d + 1)])

def truncated(s, m):
    """Continuity of the function and of its first m derivatives at every knot."""
    cols = [u(s) ** j for j in range(d + 1)]
    for k in knots:
        cols += [np.clip(u(s) - u(k), 0.0, None) ** r for r in range(m + 1, d + 1)]
    return np.column_stack(cols)

def fit(X):
    beta, *_ = np.linalg.lstsq(X, y, rcond=None)
    resid = y - X @ beta
    return np.linalg.matrix_rank(X), resid @ resid, X @ beta

for name, X in [("separate cubics", separate(year)), ("continuous", truncated(year,
0)),
               ("continuous derivative", truncated(year, 1)), ("cubic spline",
truncated(year, 2))]:
    p, rss, _ = fit(X)
    print(f"{name:24s} p = {p:2d}    residual sum of squares {rss:10.0f}")

```

[Run this code online](#)

```

from scipy import stats

p_full, rss_full, _ = fit(separate(year))
for m, label in ((0, "continuity"), (1, "first derivative"), (2, "second derivative"
)):
    p_m, rss_m, _ = fit(truncated(year, m))
    q = p_full - p_m
    F = ((rss_m - rss_full) / q) / (rss_full / (n - p_full))
    print(f"up to the {label:17s}: q = {q}, F = {F:5.2f}, p-value {1 - stats.f.cdf(F
, q, n - p_full):.4f}")

```

[Run this code online](#)

Warning. The knot at 1899 was not chosen blind: Cobb's analysis and the large quartic component in Example (A trend in the Nile) both point to the end of the nineteenth century, and both used this series. The t statistic -8.71 therefore overstates the evidence for a break of unknown date, whose null distribution is that of the largest of ninety-nine

statistics, one for each year one could split at.

Idea. Degrees of freedom are the currency, and a spline spends them differently. Each of a polynomial's $d+1$ parameters affects the curve everywhere; a spline has $d+1+K$, and the K extra act only to the right of their knots, adjusting the d th derivative. Raising K at fixed low d is a local purchase; raising d is not.

Remark (Natural boundary conditions). Outside $[\kappa_1, \kappa_K]$ a cubic spline is still a cubic, and cubics diverge. **Natural** cubic splines add $f''(a) = f'''(a) = f''(b) = f'''(b) = 0$, forcing f to be linear beyond the outer knots and reducing the dimension to K ([Exercise B1](#)) — a decision that the curve should flatten where the data run out, not a discovery about them. It becomes more than a convention in [Chapter 43](#), where the natural cubic spline exactly minimizes a penalized criterion (Theorem 43.4.2).

42.3.5 Exercises

42.3.5.1 A. Check your understanding

Exercise A1.

How many parameters has a quadratic spline with 7 interior knots? A cubic spline with 7, one of them a double knot (only C^1 there)? A piecewise constant function with 7?

Solution. By [Theorem 42.3.1](#), $2 + 1 + 7 = 10$, then $3 + 1 + 7 + 1 = 12$ (the double knot contributes $d - m = 3 - 1 = 2$ instead of 1), and $0 + 1 + 7 = 8$.

Exercise A2.

In the broken-line model $f(t) = \beta_0 + \beta_1 t + \sum_k \theta_k (t - \kappa_k)_+$, give the slope of f on (κ_j, κ_{j+1}) , the hypothesis that the line is unbent at κ_j , and the degrees of freedom of the test that it is a single straight line.

Exercise A3.

Verify that for the piecewise constant fit, changing y_i by Δ moves the fitted curve by Δ/n_k on i 's own interval and not at all elsewhere. Contrast with Proposition 42.1.1(d).

42.3.5.2 B. Practice

Exercise B1.

Show that the natural cubic splines with interior knots $\kappa_1 < \cdots < \kappa_K$ ($K \geq 2$) form a space of dimension K , by writing the four boundary conditions in the truncated power basis and checking that they are independent.

Solution. Write $f(t) = \sum_{j \leq 3} \beta_j t^j + \sum_k \theta_k (t - \kappa_k)_+^3$, with $K + 4$ coefficients. Linearity on $[a, \kappa_1]$ means $\beta_2 = \beta_3 = 0$; linearity on $[\kappa_K, b]$ makes the cubic and quadratic coefficients of the last piece vanish, which after expanding $(t - \kappa_k)^3$ reads $\sum_k \theta_k = 0$ and $\sum_k \theta_k \kappa_k = 0$. The four equations involve different coefficients and are independent, so the dimension is K .

Exercise B2.

Two straight lines meet at a known γ . Write the model in the basis $1, t, (t - \gamma)_+$, give the least squares estimate of the common value at γ and its variance, and the F statistic for equal slopes.

Solution. With $E(Y) = \beta_0 + \beta_1 t + \theta(t - \gamma)_+$, the value at γ is the estimable function $\beta_0 + \beta_1 \gamma$, estimated by $\hat{\beta}_0 + \gamma \hat{\beta}_1$ with variance $\sigma^2 \mathbf{a}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{a}$, $\mathbf{a} = (1, \gamma, 0)^\top$. Equal slopes is $\theta = 0$, tested by the square of the t statistic for $\hat{\theta}$ against $F(1, n - 3)$ (Theorem 11.6.1).

Exercise B3.

Let $f \in \mathcal{S}_d(\kappa)$ have truncated power coefficients θ_k . Show that $f^{(d)}$ is a step function jumping by $d! \theta_k$ at κ_k , and that $\theta_k = 0$ exactly when f is a single polynomial across κ_k .

Solution. Differentiating Eq. 42.3.1 d times kills every power below t^d and turns $(t - \kappa_k)_+^d$ into $d!$ times the indicator of $[\kappa_k, \infty)$, so $f^{(d)} = d! \beta_d + d! \sum_k \theta_k \mathbf{1}\{t \geq \kappa_k\}$. If $\theta_k = 0$ the pieces on either side of κ_k have the same d th derivative and agree to order $d - 1$ there, hence are the same polynomial; the converse is immediate.

42.3.5.3 C. Going deeper

Exercise C1.

Show that for $f \in C^{d+1}[a, b]$,

$$f(t) = \sum_{j=0}^d \frac{f^{(j)}(a)}{j!} (t - a)^j + \frac{1}{d!} \int_a^b f^{(d+1)}(s) (t - s)_+^d ds.$$

Interpret it as saying that the truncated power functions are a continuous basis for smooth functions, of which Eq. 42.3.1 keeps K members.

Exercise C2.

Let κ and $\kappa + \epsilon$ be two knots of a cubic spline. Show that as $\epsilon \downarrow 0$ the angle between $(t - \kappa)_+^3$ and $(t - \kappa - \epsilon)_+^3$ tends to zero, while the normalized difference $\{(t - \kappa)_+^3 - (t - \kappa - \epsilon)_+^3\} / (3\epsilon)$ tends to $(t - \kappa)_+^2$. Explain how this both shows that a double knot lowers the smoothness by one order and warns against placing knots close together in the truncated power basis.

Solution. For $t \geq \kappa + \epsilon$, $\{(t - \kappa)^3 - (t - \kappa - \epsilon)^3\} / (3\epsilon) \rightarrow (t - \kappa)^2$ by the definition of the derivative; for $t \leq \kappa$ both terms vanish, and the middle interval, of width ϵ , contributes

nothing. The limit is $(t - \kappa)_+^2$, the basis function [Theorem 42.3.1\(b\)](#) adds at a knot of multiplicity two, where only C^1 continuity is required. Two nearly coincident columns whose difference carries the information is what makes a design ill-conditioned ([Definition 1.8.1](#)), and it inflates the coefficients' variances in the sense of [Definition 26.2.1](#): the fit is fine, the coefficients are not.